

```
def binary_search(arr, key, low, high):
```

```
    if low > high:
        return -1
```

```
    mid = (low + high) // 2
```

```
    if arr[mid] == key:
        return mid
```

```
    elif key < arr[mid]:
        return binary_search(arr, key, low, mid - 1)
```

```
    else:
        return binary_search(arr, key, mid + 1, high)
```

```
arr = list(map(int, input("Enter sorted elements: ").split()))
```

```
key = int(input("Enter element to search: "))
```

```
result = binary_search(arr, key, 0, len(arr) - 1)
```

```
if result != -1:
    print("Element found at position", result + 1)
```

```
else:
    print("Element not found")
```

```
print("Time Complexity = O(log n)")
```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 2025)  
on win32

Enter "help" below or click "Help" above

>>>

==== RESTART: C:/Users/ROOPA SAI DIVYA/C

Enter sorted elements: 10 20 30 40 50

Enter element to search: 30

Element found at position 3

Time Complexity = O(log n)

>>>